DEPRESSION 1:137–142(1993)

SOCIAL FACTORS REGULATING SECURITY AND FEAR IN INFANT RHESUS MONKEYS

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Security and fear are emotional states frequently experienced by infant primates and either may be enhanced or reduced by the presence of the mother or another conspecific. These studies in rhesus monkeys examined the degree to which infants' levels of security and fear are regulated by a) the mere physical presence of the infant's mother or an unfamiliar adult male and b) the conspecific's social behavior. Infants' reactions during reunion with either their mother or an unfamiliar adult male were studied after a brief separation. In the first experiment, each infant underwent a brief maternal separation on three occasions, after which it was exposed to the following conditions: left alone for an additional period, reunited with its awake mother, and reunited with its anesthetized mother. Infants reunited with their awake mother were calm and appeared to be secure. In contrast, when reunited with their anesthetized mother they made physical contact with her but exhibited distress-related behaviors. The second experiment examined the importance of interaction in regulating fear in infants. Infants underwent brief maternal separation and then were united with an unfamiliar awake or anesthetized male. Initially, infants attempted to affiliate with the male, but in contrast to the mothers, the male discouraged physical contact and at times was aggressive. As a result, infants displayed more fear-related behaviors. When united with an anesthetized male, infants made physical contact with him and did not appear fearful. These results indicate that a prominent factor in regulating security and fear in infant rhesus monkeys is the character of their social interactions with adults and not simply the adult's physical presence. Depression 1:137–142 (1993). © *1993 Wiley-Liss, Inc.*

Key words: *security, fear, primates*

INTRODUCTION

In primates, security and fearfulness are affective states that are evident early in life and mediate important aspects of social behavior. Because security and fear are frequently reciprocal states, understanding issues regulating an infant's level of security also may help understanding factors modulating infants' level of fear.

A variety of studies in humans (Campos et al., 1983) and other primates have demonstrated the ontogeny of the expression of behaviors associated with security and fear. Infant macaques are typically dependent on their mothers during the first few months of life, when they engage in much physical contact and, when not actually in contact, maintain proximity (Harlow, 1959; Hansen, 1966; Hinde 1969; Rosenblum and Alpert, 1977). At this age infant rhesus monkeys (Harlow and Zimmerman, 1959; Kalin et al., 1991) and other macaque infants (Rosenblum and Alpert, 1977) are not likely to respond fearfully to environmental threats or strangers with overt behaviors and may even engage in exploratory responses. Around the third month, rhesus infants become increasingly independent and begin to display adult-like fear-related behaviors (Kalin et al., 1991).

There has been considerable speculation regarding the elements important in promoting security in the infant primate. Classic experiments by Harlow and coworkers suggested that "contact comfort" is one of the most important factors in enhancing security while at the same time reducing fearfulness (Harlow, 1959; Harlow and Zimmerman, 1959). In these studies, infant rhesus monkeys reared in the presence of both a non-nutritive cloth surrogate and a nutritive wire surrogate would preferentially retreat to the non-nutritive cloth surrogate when faced with a frightening situation. This experiment

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Received for publication February 1, 1993; revised March 30, 1993; accepted April 27, 1993.

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clearly demonstrates the relation between the internal states of security and fear. The infant primate reduces fear by retreating to the most physically comfortable surrogate.

While these studies made important contributions, they did not examine the elements in the interactive process between a mother and her infant that either promote or reduce security. Harlow's work showed that contact with a familiar soft and pleasant object increases security and reduces distress. Studies examining human mother-infant interactions have demonstrated that, in addition to physical contact, verbal and nonverbal cues may increase or reduce the sense of security between two individuals (Beebe et al., 1982; Brazelton et al., 1974; Campos et al., 1983; for review see Field, 1985).

In the monkey, affiliative behaviors associated with providing security include maintaining proximity, grooming, clinging, and nursing. Under appropriate circumstances, certain vocalizations may convey a desire to affiliate. The coo, which has been likened to a human cry, is a clear call that first increases then decreases in pitch (Rowell and Hinde, 1962). This vocalization is hypothesized to signal the infant's need for maternal retrieval (Rowell and Hinde, 1962; Levine et al., 1984; Levine et al., 1987). Once reunited with their mothers, infants tend to stop cooing (Kalin et al., 1992). Another vocalization thought to be affiliative is the girn (Rowell and Hinde, 1962). Girns are low, soft, nasal, whimpering sounds of short duration, frequently emitted in bouts. Recent work from our laboratory suggests that girns are used in a variety of situations and may reinforce maternal contact with the infant (Kalin et al., 1992).

Behaviors that reflect fear include grimacing, lipsmacking, submission, and shrieking. Shrieks are emitted in situations associated with extreme fearfulness and frequently function to recruit help. Another common vocalization is the bark, which is emitted by infant rhesus monkeys when they are threatened (Kalin and Shelton, 1989).

The value of studying factors mediating security and fear in nonhuman primates is that such experiments could establish social and/or physiological factors important in maintaining security in humans. Controlled prospective experiments can be more readily performed in nonhuman primates than in humans. While some studies have been performed in mammals with altricial young (Hofer and Shair, 1978, 1987), the relevance of these studies to primates may be limited, especially for understanding regulation of security and fear in human infants.

In the following studies, we examined whether the rhesus monkey mother's ability to provide social feedback is critical in determining her infant's level of security. After a brief separation, infants were reunited with their mother, who was either awake or anesthetized. To examine social factors modulating fear, a second experiment tested the hypothesis that fear induced by exposure to a strange adult male is due to the infant's interactions with that male and not simply to his presence. Fearful behaviors were compared when infants were united with either an awake or an anesthetized unfamiliar adult male.

METHOD

EXPERIMENTAL SUBJECTS

Experiment 1 used nine infant rhesus monkeys *(Macaca mulatta)* (five males and four females) and their mothers. In the second experiment eight of these mother-infant pairs were used (five males and three females). Infants ranged in age from 5 to 9.5 months (mean = 6.6 months). At this age infants still rely on their mothers for comfort but they also have developed the ability to express fearfulness in a variety of situations (Harlow, 1959; Kalin et al., 1991). Mothers ranged in age from 5 to 21 years (mean = 11.2 years) and had given birth to between zero and seven offspring prior to the infants used in these studies. Mother-infant pairs were housed at the Harlow Primate Laboratory in cages 75 cm wide × 80 cm deep × 75 cm high. Animals were maintained on a 12 h light–12 h dark cycle (lights on at 0600 h). All experimental procedures were approved by the University of Wisconsin Research Animal Resource Committee.

At approximately the same time each week, between 0830 and 1330 h, each infant was separated from its mother. During separation the infant was placed in a cage 150 cm wide × 80 cm deep × 78 cm high in a different room, where it remained for 20 min in auditory and visual isolation from other monkeys. In the first experiment infants were exposed to a different test condition each week in a counterbalanced fashion. The three test conditions were:

1. *Alone.* The infant was removed from the separation cage and placed in another cage by itself for 9 min.
2. *Anesthetized mother.* After separation the infant was reunited for 9 min with its mother that had been anesthetized with ketamine HCl (10 mg/kg i.m.).
3. *Awake mother.* The infant was reunited with its awake mother for 9 min.

The second experiment followed the same design. However, instead of being reunited with their mothers, infants were placed in a cage with an unfamiliar 5-year-old adult male. To reduce the variability in the behavioral interaction between the male and infants, we used the same male for all testing. He had been raised with his mother for his first year and then socially housed in a group-living situation until he was 4.5 years old. During the 6 months preceding the current study, he lived in single-caged housing. This particular male was selected because of his behavioral history with other infants, suggesting that he was unlikely to cause serious harm to them. Like the mother, the male was either anesthetized or awake. In this experiment we have data for only eight of the infants used in experiment 1 because the male attacked one infant when it was placed with him. The infant was immediately removed and dropped from the study.

Behaviors and vocalizations were scored from videotapes by a single trained rater using a scoring system developed in our laboratory (Kalin and Shelton, 1989). Interactive behaviors were scored with the infant as the focal animal. The behaviors of interest are defined in Table 1. Vocalization data were collected as frequency of

TABLE 1. Behavioral and vocal categories

Category	Definition
Behavior	
Locomotion	Movement of one or more steps at any speed. Includes dropping from ceiling to floor or swinging.
Ventral contact	Gross body contact between the ventrum of an infant with either the mother or the male.
Nonventral contact	Gross body contact of infant with nonventral part of male or mother's body.
Aggression	Infant receives biting, vigorous hair-pulling, or other gestures such as being pushed away.
Submission	Oriented withdrawal, fear grimacing, cowering, fleeing, or yielding space.
Vocalizations[a]	
Bark	Short, rasping, low-frequency sound emitted with mouth held loosely open.
Coo	Long-duration call with initial rise then fall in frequency. Lips rounded and pursed.
Girn	Low-frequency soft nasal whine. Lips closed or slightly open. Often given in bouts.
Shriek	Harsh, brief, high-pitched sound with moderate frequency modulation. Lips are pulled back.

[a]*Sonograms and/or analyses can be found in Kalin, et al. (1988, 1992).*

occurrence during three 3-min periods and normalized using a square root transformation. Other behaviors, scored as durations of occurrence during the three 3-min periods, were normalized with a logarithmic transformation. Repeated measures ANOVAs were performed using Duncan's Multiple Range Test for post hoc comparisons. The alone condition was exluded from analysis of interactive behaviors. Nonparametric analyses were performed when appropriate: McNemar's test for the comparison of two related samples and Cochran's Q test for comparison of three related samples.

Although experiments 1 and 2 were performed separately, we also analyzed the behaviors of interest comparing effects of the conditions in the two studies. In these analyses, data from the eight animals that completed both studies were used. Data from the alone conditions in the two experiments were averaged and compared with data for the anesthetized mother, awake mother, anesthetized male, and awake male conditions.

RESULTS

EXPERIMENT 1: REUNION WITH THE MOTHER

Behaviors. Awake mothers engaged in behavior directed at their infants, such as holding and looking at them, which anesthetized mothers could not engage in. Even though mothers were anesthetized, infants still sought ventral contact with them and the number of mother-infant pairs engaging in ventral contact did not significantly differ whether the mother was awake (9/9) or anesthetized (6/9). However, duration of ventral contact was much greater when infants were with their awake mothers (mean ± SEM = 160.2 ± 18.9 s) than when with their anesthetized mothers (mean ± SEM = 23.8 ± 12.8 s) [$F(1,8) = 19.43$; $p < .003$)]. When with their anesthetized mothers, 8/9 infants engaged in nonventral contact. Infants never sought nonventral contact when reunited with their awake mother ($\chi^2 = 6.12$; d.f. = 1; $p < .02$). Additionally, infants united with their anesthetized mothers engaged in significantly more nonventral contact as the period of reunion progressed [$F(2,16) = 4.14$; $p < .04$]. Infants' level of locomotion significantly differed across the three test conditions: alone > anesthetized mother > awake mother [$F(2,16) = 17.1$; $p < .000001$]. These differences are inversely related to the amount of time infants engaged in ventral contact with their mothers. (In the alone condition infants did not have this option.)

Vocalizations. When infants were reunited with their awake mother, significantly fewer infants emitted coos ($Q = 7.0$; d.f. = 2; $p < .05$). Six of nine infants cooed when alone and seven of nine cooed when with their anesthetized mother. In contrast, only two of nine infants cooed when reunited with their awake mother. ANOVA revealed that test conditions also affected the number of coos [$F(2,16) = 7.89$; $p < .004$]. As can be seen in Figure 1, the relative numbers of coos emitted by infants in the different conditions were anesthetized > alone = awake.

Girns were emitted by only one of nine infants in the alone condition, compared to seven of eight and eight of nine when infants were reunited with their anesthetized and awake mothers, respectively ($Q = 10.8$; d.f. = 2; $p < .01$). Main effects of test condition [$F(2,16) = 7.44$; $p < .006$] and time [$F(2,16) = 15.57$; $p < .00001$] were also observed. There was a significant test condition × time interaction [$F(4,32) = 7.42$; $p < .0002$] (Fig. 2). Post-hoc testing revealed that significantly more girns were emitted by infants when with their mothers, either awake or



Figure 1. Number of coos (square root transformation, mean ± SEM) emitted per 9 min when the infant remained alone was reunited with its anesthetized mother and was reunited with its awake mother (N = 9). *p <.05, anesthetized mother condition > other conditions.

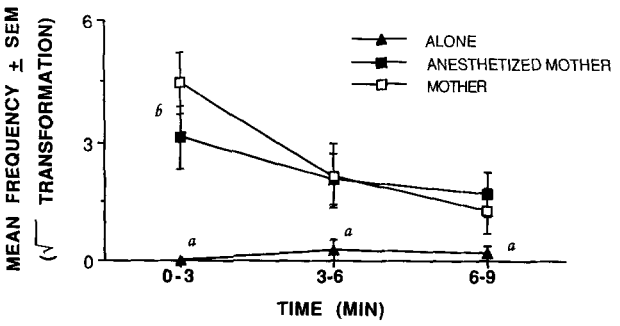


Figure 2. Effects of test conditions over time (three successive 3-min periods) on the frequency of girns (square root transformation, mean ± S.E.M.) emitted by infants (N = 9). (a) All alone time periods differ from all reunion time periods, $p < .05$. (b) During the first 3 min, more girns occurred with the awake mother than with the anesthetized mother, $p < .01$.

anesthetized, than by infants remaining alone across all three time periods. The only difference observed between infants reunited with their awake compared to their anesthetized mothers occurred during the first three minutes, when infants initially emitted significantly more girns when reunited with their awake mothers.

Barks were emitted at low levels and were also significantly affected by test conditions ($Q = 10.1$; d.f. = 2; $p < .01$). Infants never barked when reunited with their awake mothers. In contrast, six of nine infants barked when alone and when reunited with their anesthetized mothers. Shrieks rarely occurred and neither the number of infants that shrieked nor the frequency of shrieks differed significantly between test conditions.

EXPERIMENT 2: EFFECTS OF AN UNFAMILIAR ADULT MALE

Behaviors. The awake male socially explored all eight infants and directed brief bouts of mild aggression toward four of them. Typically he would push the infant to the floor of the cage while mouthing its body parts and pulling its skin with his teeth. These incidents did not cause physical harm to the infants. Seven of the eight infants engaged in submissive behaviors in the presence of the awake male, whereas none of the infants were submissive in the presence of the anesthetized male ($\chi^2 = 5.2$; d.f. = 1; $p < .05$). Only two of the male-infant pairs engaged in ventral contact (one infant with the awake and one with the anesthetized male), and in both cases this was of very brief duration. In contrast, nonventral contact occurred in six of eight infants in the presence of the anesthetized male compared to zero of eight infants when with the awake male ($\chi^2 = 4.2$; d.f. = 1; $p < .05$).

Vocalizations. Shriek was the only vocalization that significantly differed with test conditions. A greater number of shrieks were emitted by infants when united with the awake male than with the anesthetized male and when in the alone condition [$F(2,14) = 4.11$; $p < .04$] (Fig. 3). Girns occurred at a low rate and significantly decreased over time in all test conditions [$F(2,14) = 8.61$; $p < .004$].

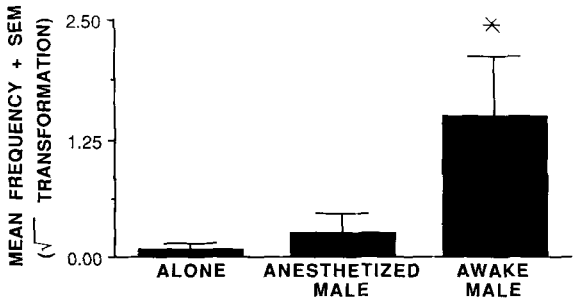


Figure 3. Effects of unfamiliar adult male on frequency of infants' shrieking per 9 min (square root transformation, mean ± S.E.M.) (N = 8). *$p < .05$, awake male > anesthetized male and alone conditions.

INFANTS' BEHAVIOR IN PRESENCE OF MOTHER VS. IN PRESENCE OF MALE

Significant main effects of condition were observed for ventral [$F(3,21) = 29.17$; $p < .000001$] and nonventral [$F(3,21) = 17.05$; $p < .00001$] contact. More ventral contact occurred with the awake mother than in all other conditions. While less ventral contact occurred with the anesthetized mother, this amount was significantly more than occurred with the anesthetized male. Nonventral contact occurred only in the anesthetized conditions and occurred in greater amounts when infants were with their mothers than when with the anesthetized male [$F(1,7) = 6.70$; $p < .04$]. A significant condition × time interaction was observed. As can be seen in Figure 4, nonventral contact increased over time with the anesthetized mother but not with the anesthetized male [$F(2,14) = 4.44$; $p < .04$].

The number of girns emitted also differed with test condition [$F(4,28) = 6.98$; $p < .0005$]. More girns occurred in both conditions with the mother than in all other conditions. A test condition × time interaction also

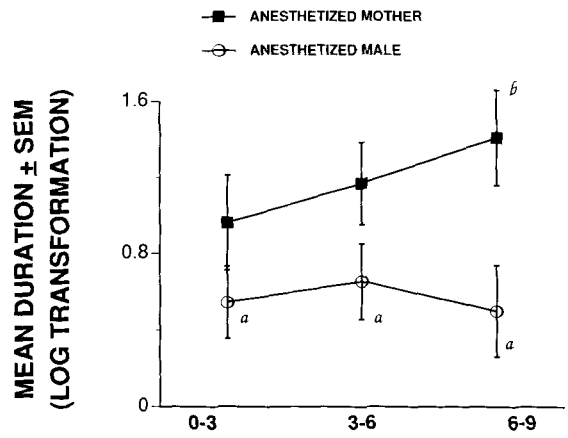


Figure 4. Duration of infant nonventral contact (log transformation, mean ± S.E.M.) occurring over time (three 3-min test periods) with the anesthetized mother and the anesthetized male (N = 8). (a) $p < .01$, anesthetized mother condition > anesthetized male condition. (b) $p < .01$, last 3-min period with anesthetized mother > first 3-min period with her.

occurred [$F(8,56) = 3.74$; $p < .002$]. While at different levels, girns significantly decreased over time in the awake and anesthetized mother and male conditions. Girning did not change over time when infants were alone. Significantly more coos were emitted during the anesthetized mother, anesthetized male, and awake male conditions than during the awake mother condition. Infants in the awake mother and alone conditions did not differ significantly in the number of coos they emitted [$F(4,28) = 5.44$; $p < .003$]. Shrieks were also significantly affected by test conditions [$F(4,28) = 2.79$; $p < .046$]. Infants shrieked the most in the awake male condition. This was significantly greater than in the alone, awake mother, and anesthetized male conditions ($ps < .05$). Infants' shrieking in the male's presence was not significantly greater than with the anesthetized mother.

DISCUSSION

After an infant reaches 2 months of age a dramatic decrease occurs in rhesus mother-infant ventral contact (Berman, 1980; Hansen, 1966). However, after this age, reunion following separation results in increased levels of maternal-infant contact similar to those seen in very young undisturbed infants (Seay et al., 1962; Hinde and Davies, 1972; Hinde and White, 1974). As expected, we found that after a brief separation, 5- to 9.5-month-old infants spent much time in ventral contact with their mother. This was associated with reduced levels of cooing and barking. In the first experiment, fewer infants emitted coos when reunited with their awake mother. A nonsignificant reduction in the number of coos also occurred in this condition compared to when infants were alone.

Little girning occurred when infants were alone, but during the first 3 min of reunion with the awake mother, infants emitted significantly more girns than they emitted in the presence of the anesthetized mother. These findings confirm results of our earlier studies demonstrating differences in the infant's expression of coos and girns (Kalin et al., 1992). Since coos are reduced by reunion with the awake mother and girns increase with the awake mother, it appears that coos function to establish proximity to the mother, whereas girns may play an interactive role to maintain contact with her.

A rapid reduction in an infant's level of distress is mediated by reunion with the awake mother. In contrast to the calming effect of the awake mother, reunion with the anesthetized mother increased the level of infants' distress. These infants cooed and barked more than when they were with their awake mothers. They were motivated to gain contact with their mother; as they engaged in some nonventral contact and over the period of testing, nonventral contact increased. However, this physical contact was not sufficient to reduce their distress.

Studies with human infants suggest that verbal and nonverbal communication are very important in a mother's ability to provide security to her infant. In addition to talking, nonverbal gestures such as glancing, facial expressions, and touching are thought to be important in the human infant's perception of safety and security. Field et al. (1990) have suggested that the human mother behaves so as to modulate her infant's optimal level of stimulation and arousal. Experiments in which human mothers maintain a still, nonreactive face in their infants' presence increase the level of the infants' distress (Gusella and Muir, 1988). Our paradigm in which the infant is presented with its unreactive mother may be analogous to the still-face mother paradigm. This raises the possibility that the infants' distress with the anesthetized mother is not simply due to a lack of comfort supplied by her but is related to her inability to interact in her usual predictable way. These alternatives are not mutually exclusive and both may be important in enhancing infants' security.

These results demonstrate a fundamental difference between primates and rats. In rats, a reduction in both isolation-induced ultrasonic distress vocalizations (Hofer and Shair, 1978, 1987) and pituitary-adrenal activation (Stanton et al., 1987) occurs when isolated pups are exposed to their anesthetized mother. This occurs even when pups are unable to suckle from the anesthetized dam.

In contrast to the reduced distress shown upon reunion with their mother, infants responded to the unfamiliar male with fear. The male thwarted any affiliative attempts and engaged in short aggressive bouts. These were quite effective in discouraging further affiliative attempts by the infant and elicited shrieking and submissive behaviors. Since these frightened infants would benefit from their mother's protection, it is interesting that they did not coo more when with the awake male. The relative lack of cooing could be due to the increased shrieking, which reflects an extreme state of distress and also serves to recruit help from conspecifics. When infants were with the anesthetized male their behavior changed dramatically. They engaged in nonventral contact with the male and did not display the level of fear seen with the awake male. This experiment demonstrates that the infants' perception of the male as a significant threat was related to their behavioral interaction with him and not simply to the fact that he was an unfamiliar male.

We compared differences in the magnitude and pattern of responses between infants in the presence of their mother and of the male. When infants were with the anesthetized male, their behavior indicated that their motivation to achieve physical contact was not as great as when they were with their anesthetized mother. Infants never engaged in ventral contact with the anesthetized male but did so with their anesthetized mothers. Infants engaged in nonventral contact with the anesthetized male but at lower levels than with the anesthetized mother; over time an increase in nonventral contact occurred with infants in the presence of their anesthetized mother, but this was not seen when infants were with the anesthetized male.

Girning occurred more with the awake mother than in all other conditions, and cooing was significantly reduced only when infants were with their awake mother. Infants shrieked as much when with their anesthetized

mother as when in the presence of the anesthetized male. The cooing and shrieking data again suggest that even though infants achieved physical contact with their anesthetized mother they remained distressed.

These data indicate that the quality of behavioral interaction and not the mere physical presence of a conspecific modulates the primate infant's level of security or fear. Future experiments using this paradigm will identify the specific interactive elements that are important and examine physiological factors that regulate primate infants' security and fear.

Acknowledgments. This work was supported in part by the Medical Research Service, U.S. Department of Veterans Affairs, and by NIMH grant 1 RO1 MH-46729 to NHK and NIMH Research Scientist Award MH-00177 to CTS. The authors appreciate the technical assistance of Helen Van Valkenberg and the editorial assistance of Carol Steinhart.

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